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ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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Automated content categorization improve
email classification accuracy and organization
compared to traditional email classification

ABSTRACT

1. Email management is challenging due to the increasing number of emails, making accurate classification essential for efficient organization.
2. This project uses **Artificial Intelligence (AI)**, **Machine Learning (ML)**, and **Natural Language Processing (NLP)** to automatically categorize emails based on their content.
3. The system classifies emails into categories such as **Spam, Promotions, Social, Work, Personal, and Updates**, reducing manual effort and improving organization.
4. Techniques such as **text preprocessing**, **TF-IDF feature extraction**, and **machine learning algorithms** (e.g., Naïve Bayes or Logistic Regression) are used to improve classification accuracy.
5. The proposed automated content categorization system achieves better **accuracy, efficiency, and email organization** compared to traditional email classification methods, helping users manage emails more effectively.

Introduction

- 1. Email is a primary channel for customer service, operations, finance, and internal communication.**
2. Organizations receive a continuous stream of messages with different intents and priorities.
3. Manual reading, labeling, and forwarding slow down the response process.
4. AI-based classification can identify the purpose of a message and recommend the next action.
- 5. The goal is faster routing, consistent decisions, and meaningful human oversight.**

Objective

1. **To design a reliable framework that classifies organizational emails and supports timely, explainable action.**

Identify the intent, topic, priority, and actionability of incoming email.

2. Reduce repetitive manual triage while preserving human review for uncertain cases.
3. Improve routing consistency, response speed, and visibility into recurring demand.
4. Define safeguards for privacy, fairness, traceability, and responsible automation.

Methodology

1. Collect representative email samples and define a clear organizational taxonomy.
2. Clean text, metadata, and thread context while removing unnecessary sensitive content.
3. Apply rules or machine-learning models to predict labels with confidence scores.
4. Evaluate precision, recall, coverage, review rate, and routing time.
5. **Use human corrections as controlled feedback for improvement.**

MODULE 1: Email Data Collection

1. Collect emails from organizational email systems and communication platforms.
2. Extract important information such as subject, sender, body, and attachments.
3. Remove duplicate, spam, and irrelevant emails to improve data quality.
4. Clean and preprocess text by removing special characters and unnecessary words.
5. Protect confidential information using data masking and privacy techniques.
6. Prepare structured and standardized data for AI model training.

MODULE 2: Email Classification

1. Analyze email content using Natural Language Processing (NLP) techniques.
2. Identify the email category such as Support, Finance, HR, Sales, or General Inquiry.
3. Determine the priority level based on urgency and business requirements.
4. Generate confidence scores for each prediction to improve decision-making.
5. Enhance classification accuracy through machine learning algorithms.
6. Continuously improve the model using new labeled email datasets

MODULE 3:Email Management

1. Automatically forward classified emails to the appropriate department or user.
2. Allow manual verification for emails with low confidence scores.
3. Monitor system performance using accuracy, precision, recall, and F1-score.
4. Maintain detailed logs for tracking and auditing email processing activities.
5. Collect user feedback to retrain and improve the AI classification model.
6. Ensure secure, reliable, and efficient email management across the organization.

LITERATURE REVIEW

Author(s)	Year	Research Title	Method Used	Key Findings
Salton & McGill	1983	Introduction to Modern Information Retrieval	Vector Space Model	Introduced text representation techniques for document classification.
Joachims	1998	Text Categorization with Support Vector Machines	Support Vector Machine (SVM)	Demonstrated high accuracy in email and text classification tasks.
Sebastiani	2002	Machine Learning in Automated Text Categorization	Machine Learning Algorithms	Reviewed various text classification methods and evaluation techniques.
Vaswani et al.	2017	Attention Is All You Need	Transformer Architecture	Introduced the Transformer model, improving natural language understanding.
Devlin et al.	2019	BERT: Pre-training of Deep Bidirectional Transformers	BERT (Deep Learning)	Achieved state-of-the-art performance in text and email classification.

Problem Statements

1. Organizations receive a large number of emails every day, making manual sorting difficult
2. Important emails may be delayed or misplaced due to incorrect manual classification.
3. Existing keyword-based filtering methods cannot accurately understand email context.
4. Manual email management increases workload, response time, and operational costs.
5. There is a need for an AI-based system that can classify emails accurately and efficiently.
6. The system should improve productivity by automatically organizing and routing emails to the appropriate department.

Conclusion

1. AI-based email classification improves the accuracy and efficiency of email management.
2. Automatic classification reduces manual effort and saves valuable time.
3. Intelligent routing ensures emails reach the appropriate department quickly.
4. Natural Language Processing enhances the understanding of email content and intent.
5. The proposed system increases productivity and supports effective organizational communication.